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Optimal maintenance policy considering maintenance errors for systems operating under performance-based contracts

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ABSTRACT

In recent years, performance-based contracts (PBCs) have been extensively adopted in both manufacturing and service industries. However, maintenance strategies under PBCs are not often addressed. This paper considers a novel maintenance policy for a single-component system operating under a PBC. Failures of this system may either be hard or soft, but can only be revealed by periodic inspection. Imperfect repair (subject to the virtual age model) is used to fix a soft failure. However, imperfect repair actions may be ineffective and could cause a catastrophic failure (maintenance error) that ultimately leads to replacement. In the model proposed in this paper, we consider maintenance error, and the system is replaced upon a hard failure, a maintenance error, or preventively replaced at the nth soft failure. The objective of this model is to maximize the expected net revenue of the supplier operating under a PBC. Moreover, we compare the properties of the optimal maintenance policy under three different failure rate functions and show that, with different parameters, our model reduces to several classic maintenance models. Finally, a case study on a wind turbine system is provided to illustrate the application of the maintenance model.

1. Introduction

In the past few decades, performance-based contracts (PBCs) have attracted considerable attention in both commercial and military fields (Lin, Lin, & Yeh, 2013; Mirzahosseinian & Piplani, 2011; Ng, Maull, & Yip, 2009; Selviaridis & Wynstra, 2015). A PBC is designed around the key performance measures of customers, and it is the responsibility of suppliers to achieve higher performance by taking effective maintenance actions. The main difference between a PBC and a traditional contract is that the suppliers are rewarded based on system performance. For example, when a power engine system is operating under a PBC, the company is paid based on the availability or operating hours of the engine rather than the cost of labour and spare parts.

A PBC is likely to improve system performance indices such as availability and reliability at a lower cost. Therefore, it is of practical value to investigate the optimal maintenance policy within a PBC context. However, most existing studies on PBCs analyze the structure of a PBC or define frameworks for its implementation (Berkowitz, Gupta, Simpson, & Mcwilliams, 2005; Devries, 2004). To the authors' best knowledge, maintenance strategies under PBCs are rarely addressed. Mirzahosseinian and Piplani (2011) studied a repairable parts inventory system operating under a PBC. In addition, Jin, Ding, Guo, and Nalajala (2012) investigated the analytical relationship between system cost, reliability and spare parts stocking. An interesting problem that has not been considered in the literature is the optimal maintenance policy from the perspective of an equipment supplier. To fill this gap, we present a quantitative model to determine the optimal maintenance policy that maximizes the expected net revenue of an equipment supplier.

In practical applications, maintenance error is one of the primary causes of accidents (Caputo, Pelagagge, & Salini, 2017; Peng, 2014; Rankin, 2000). For one thing, system configurations are becoming more complex, requiring more frequent repair activities with different restoration effects. In these systems, improper repair actions could have severe consequences, resulting in immediate system replacement. Moreover, because the level of performance degradation varies when different system failures occur, a maintenance action could be less effective if it is incorrect action. Nevertheless, we find that maintenance errors have not been considered in most existing maintenance models, including the minimal repair model (Gupta, De, & Chatterjee, 2017; Lu, Zhou, & Li, 2016; Xiao & Peng, 2014), imperfect repair model (Doyen &

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Nomenclature

PBC	performance-based contract
WTS	wind turbine system
a	fixed value of a contract
b	coefficient of incentive in a PBC
A	system steady-state availability
$A_{\min}$	the minimum steady-state availability the contractor needs to achieve
X	lifetime of the system
$F(x)$	cumulative distribution function of X
T	inspection interval
n	number of soft failures that occur when the system is preventively replaced
p_s	conditional probability of a soft failure given a failure occurrence
$1-p_s$	conditional probability of a hard failure given a failure occurrence
p_m	conditional probability of an imperfect repair given a soft failure occurrence
$1-p_m$	conditional probability of a maintenance error given a soft failure occurrence
U	total uptime or operating time in a renewal cycle
S_k	total uptime of the system when k system failures have occurred
$f_{S_k}(t)$	density function of S_k
$\lambda(t)$	failure rate of the system at time t
$H(t)$	cumulative failure rate of the system at time t
N_f	number of system failures in a renewal cycle
N_I	number of inspections in a renewal cycle
N_I^i	number of inspections between the $(i-1)$th failure and the ith failure
X_i	uptime between the $(i-1)$th system failure and the ith system failure
N_k	number of inspections when k system failures have occurred
$A(T,n)$	steady-state availability as a function of T and n
c_I	cost of an inspection
c_r	cost of a repair
$\tilde{c}_r$	cost of a replacement
$R(T,n)$	expected net revenue of the supplier as a function of T and n

Gaudoin, 2004; Khatab, Ait-Kadi, & Rezg, 2014; Zhao & Xie, 2017), and perfect repair model (Golmakani & Moakedi, 2012; Qiu, Cui, & Gao, 2017). Furthermore, the effects of maintenance errors under a PBC have never been studied. This motivates us to develop a novel maintenance model, where the effects of maintenance errors are incorporated into the maintenance model of a system operating under a PBC.

As the system configuration and failure mechanisms of units become more diverse, the reliability and maintenance modelling of systems with multiple failure modes draw increasing attention. For different failure modes, the corresponding repair activities and cost implications may differ. Generally, system failure modes can be classified into two categories based on their consequences: soft failures and hard failures (Cha, Lee, & Mi, 2004; Levitin, Zhang, & Xie, 2006; Taghipour, Banjevic, & Jardine, 2010; Zheng, Zhou, Zheng, & Wu, 2016). A soft failure is usually a subcritical failure and an imperfect repair is a common maintenance response. A hard failure is a critical failure and replacement is a straightforward solution. In this paper, we model the maintenance of a complex system that is subject to both soft and hard failures.

In many existing maintenance models, failures are assumed to be detected simultaneously. However, for systems suffering from hidden failures, inspection is a crucial maintenance action that improves system availability by reducing the amount of system downtime due to failures (Cui & Xie, 2001, 2005; Lin et al., 2013; Liu et al., 2013; Peng, Feng, & Coit, 2010). Based on intervals between successive inspections, inspection policies in the literature can be further divided into two categories: periodic and non-periodic inspections. In industry, periodic inspections are more popular because of their convenience.

Within a PBC framework, this paper considers a periodically inspected repairable system under imperfect repair. The repair efficiency is characterized by a type I Kijima virtual age model (Kijima, 1989). The system is replaced in the following three scenarios, depending on whichever occurs first: (i) a hard failure; (ii) a maintenance error; (iii) the nth$(n=1,2,\ldots)$ soft failure. In contrast with existing maintenance models (Cheng & Li, 2012; Sheu, Li, & Chang, 2012; Sheu, Tsai, Wang, & Zhang, 2015), we consider the case where replacements may be induced by maintenance errors. The proposed model can be converted into various maintenance models through different parameter settings.

The rest of this paper is organized as follows. Section 2 describes the maintenance optimization problem of a system operating under a PBC while considering maintenance errors. Section 3 obtains explicit analytical expressions of the expected net revenue using the Kijima virtual age model. The properties of the optimal solutions are discussed in Section 4. Numerical examples considering WTS under a PBC are given in Section 5. The conclusion and future research directions are presented in Section 6.

2. Problem description

In this paper, an inspection-based maintenance model is developed for systems operating under a PBC. During the repair process, maintenance errors caused by improper repair actions, which can potentially trigger catastrophic failures, are taken into consideration. The basic assumptions of this model are listed as follows.

Assumption 1. Steady-state availability A is the performance measurement and the minimum steady-state availability the contractor needs to achieve is $A_{\min}$. Moreover, the expected net revenue function is defined as (Nowicki, Kumar, Steudel, & Verma, 2008)

$$R = a + b\times(A - A_{\min}) - E(C), \tag{1}$$

where a denotes the fixed value of the contract, $b\times(A-A_{\min})$ represents the performance-based incentive and $E(C)$ is the expected maintenance cost over a renewal cycle.

Assumption 2. The random time to failure is denoted by X, which has distribution function $F_X(x)$. There are failure modes of the system: soft failure, which occurs with probability p_s, and hard failure, which occurs with probability $1-p_s$.

Assumption 3. System failures can only be detected through inspections, which are carried out every T time units after a system is put into operation. The cost of an inspection is c_I.

Assumption 4. Maintenance actions are carried out when a soft failure is detected, which imperfectly repair the system with probability p_m or cause a catastrophic system failure with probability

$1-p_m$ (i.e., maintenance error occurs). The Kijima virtual age model is used to describe the effect of an imperfect repair. The cost of a repair is c_r.

Assumption 5. The system is preventively replaced at the nth soft failure or correctively replaced upon a hard failure or a maintenance error, whichever occurs first. The cost of such a replacement is $\tilde{c}_r$.

A renewal cycle is defined as the time interval between the installation of a new system and the first replacement or a time interval between two consecutive replacements. Hence, the maintenance cost in a renewal cycle is associated with inspections, repairs and replacement. Our objective is to find the optimal inspection interval that maximizes the expected net revenue of the supplier. We then derive the expected net revenue of the supplier.

3. Expected net revenue of suppliers

3.1. System steady-state availability

3.1.1. Expected system uptime over a renewal cycle

According to Assumptions 2 and 4, when a failure occurs, the type of system failure and the corresponding maintenance action may vary. Let $Y_i, (i=1,2,\dots,n)$ denote the type of the ith system failure and its corresponding maintenance result. Then the values of Y_i are defined as follows

$$Y_i=\begin{cases}0, & \text{the system fails from a hard failure}\\ 1, & \text{the system fails from a soft failure and is imperfectly repaired}\\ 2, & \text{the system fails from a soft failure and a maintenance error occurs}\end{cases}$$

The probability mass function of $Y_i, (i=1,2,\dots,n)$ is given by

$$P(Y_i=y)=\begin{cases}1-p_s, & y=0\\ p_sp_m, & y=1\\ p_s(1-p_m), & y=2\end{cases} \tag{2}$$

Let N_f represent the number of system failures in a renewal cycle and $X_i(i=1,2,\dots,N_f)$ be the uptime between the $(i-1)$th and the ith system failures. The total uptime in a renewal cycle is expressed by

$$U=X_1+X_2+\cdots+X_{N_f}. \tag{3}$$

N_f is an integer up to n. For $k=n$, $\{N_f=k\}$ implies that the system is replaced at the nth system failure before a hard failure or a maintenance error occurs. Its probability is given by

$$P\{N_f=n\}=P\{Y_1=1\cap Y_2=1\cap\cdots\cap Y_{n-1}=1\}=(p_sp_m)^{n-1}. \tag{4}$$

For $k\in(1,2,\dots,n-1)$, $\{N_f=k\}$ implies that the preceding $k-1$ failures are soft and imperfectly repaired, and the system is replaced at the kth failure. Using the total probability decomposition method, the probability of $\{N_f=k\}$ is expressed as

$$\begin{aligned}P\{N_f=k\}&=P\{Y_1=1\cap Y_2=1\cap\cdots\cap Y_k=0\}+P\{Y_1\\&=1\cap Y_2=1\cap\cdots\cap Y_k=2\}.\end{aligned} \tag{5}$$

The first term of (5) corresponds to a replacement caused by a hard failure while the second term represents a maintenance error. Then (5) can then be derived as

$$\begin{aligned}P\{N_f=k\}&=(p_sp_m)^{k-1}(1-p_s)+(p_sp_m)^{k-1}p_s(1-p_m)\\&=(p_sp_m)^{k-1}(1-p_sp_m).\end{aligned} \tag{6}$$

Using the results derived in (4) and (6), the probability mass function of N_f is given by

$$P\{N_f=k\}=I_{k<n}(p_sp_m)^{k-1}(1-p_sp_m)+I_{k=n}(p_sp_m)^{k-1}, \tag{7}$$

where I is the indicator function, which equals 1 if its argument is true and 0 otherwise.

From (7), the uptime in a renewal cycle defined in (3) is expressed as

$$U=\begin{cases}X_1, & \text{with probability } (1-p_sp_m),\\ X_1+X_2, & \text{with probability } (p_sp_m)(1-p_sp_m),\\ \dots & \\ X_1+X_2+\cdots+X_k, & \text{with probability } (p_sp_m)^{k-1}(1-p_sp_m),\\ \dots & \\ X_1+X_2+\cdots+X_{n-1}+X_n, & \text{with probability } (p_sp_m)^{n-1}.\end{cases} \tag{8}$$

Suppose that $S_k=\sum_{i=1}^{k}X_i$ represents the total uptime of the system when k system failures have occurred in a renewal cycle. Based on the virtual age model, after the kth repair, virtual age S_k becomes $S_k=S_{k-1}+aX_k$. When $a=0$, the system age is zero after each repair, which implies that a perfect repair has been performed. A minimal repair is denoted by $a=1$. If $0<a<1$, the system is maintained by imperfect repair and is younger after each repair. From Kijima (1989), the density function of S_k can be recursively given by

$$f_{S_k}(t)=\begin{cases}f(t), & k=1,\\ \frac{1}{a}\int_0^t\frac{f\left(y+\frac{t-y}{a}\right)}{R(y)}f_{S_{k-1}}(y)dy, & k\geqslant 2.\end{cases} \tag{9}$$

Based on (8) and (9), $E(U)$, the expected uptime in a renewal cycle is given by

$$\begin{aligned}E(U)&=\sum_{k=1}^{n-1}E(S_k)P(N_f=k)+E(S_n)P(N_f=n)=(1-p_sp_m)\\&\quad\left(\sum_{k=1}^{n-1}(p_sp_m)^{k-1}\int_0^\infty tf_{S_k}(t)dt\right)+(p_sp_m)^{n-1}\int_0^\infty tf_{S_n}(t)dt.\end{aligned} \tag{10}$$

Remark. Let $\lambda(t)=f(t)/R(t)$ be the failure rate function of the system at time t and $H(t)=\int_0^t\lambda(u)du$ represent the cumulative hazard rate function. For the special case of minimal repair, $f_{S_k}(t)$ in (9) is expressed as

$$f_{S_k}(t)=\frac{f(t)[H(t)]^{k-1}}{(k-1)!}. \tag{11}$$

Substituting (11) into (10), the expected uptime is given by

$$\begin{aligned}E(U)&=(1-p_sp_m)\left(\sum_{k=1}^{n-1}(p_sp_m)^{k-1}\int_0^\infty\frac{tf(t)[H(t)]^{k-1}}{(k-1)!}dt\right)\\&\quad+(p_sp_m)^{n-1}\int_0^\infty\frac{tf(t)[H(t)]^{n-1}}{(n-1)!}dt.\end{aligned} \tag{12}$$

To be specific, for an exponential distribution with failure rate λ, we have

$$E(U)=\frac{1-(p_sp_m)^{n-1}+(1-p_sp_m)(p_sp_m)^{n-1}}{\lambda(1-p_sp_m)}. \tag{13}$$

The detailed derivation of (13) is given in Appendix A.

3.1.2. Expected length of a renewal cycle

Let N_I represent the number of inspections in a renewal cycle. Moreover, $N_I^i, (i=1,2,\dots,n)$ denotes the number of inspections between the $(i-1)$th and ith failures. Then N_I can be given by

$$N_I=N_I^1+N_I^2+\cdots+N_I^{N_f}. \tag{14}$$

Using the results in (7) and (14), N_I can be expressed as

$$N_I = \begin{cases} N_I^1, & \text{with probability } (1-p_sp_m), \\ N_I^1 + N_I^2, & \text{with probability } (p_sp_m)(1-p_sp_m), \\ \dots & \\ N_I^1 + N_I^2 + \cdots + N_I^k, & \text{with probability } (p_sp_m)^{k-1}(1-p_sp_m), \\ \dots & \\ N_I^1 + N_I^2 + \cdots + N_I^{n-1} + N_I^n, & \text{with probability } (p_sp_m)^{n-1}. \end{cases} \tag{15}$$

Because N_I^i is related to X_i in a way that $(N_I^i - 1)T < X_i < N_I^i T$, the probability mass function of N_I^i is given by

$$P(N_I^i = m) = F_i(mT) - F_i((m-1)T), \tag{16}$$

where $F_i(t)$ denotes the distribution function of X_i, which can be obtained from Kijima (1989) as follows:

$$F_i(t) = 1 - \int_0^\infty \frac{F(t+x) - F(x)}{R(x)} f_{S_{i-1}}(x)dx. \tag{17}$$

Suppose that $N_k = \sum_{i=1}^k N_I^i$ represents the number of inspections when a renewal cycle includes k system failures. Then, the expected number of inspections in a renewal cycle, is expressed as

$$E(N_I) = \sum_{k=1}^{n-1}\sum_{i=1}^{k}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_sp_m)^{k-1}(1-p_sp_m) + \sum_{i=1}^{n}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_sp_m)^{n-1}. \tag{18}$$

The detailed derivation of (18) is given in Appendix A. The expected length of a renewal cycle is therefore

$$E(L) = T\left[\sum_{k=1}^{n-1}\sum_{i=1}^{k}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_sp_m)^{k-1}(1-p_sp_m) + \sum_{i=1}^{n}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_sp_m)^{n-1}\right]. \tag{19}$$

According to the renewal theorem (Ross, 1996), the steady-state availability of the system as a function of T and n can be obtained as follows

$$A(T,n) = \frac{E(U)}{E(L)} = \frac{(1-p_sp_m)\left(\sum_{k=1}^{n-1}(p_sp_m)^{k-1}\int_0^\infty tf_{S_k}(t)dt\right) + (p_sp_m)^{n-1}\int_0^\infty tf_{S_n}(t)dt}{T\left[\sum_{k=1}^{n-1}\sum_{i=1}^{k}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_sp_m)^{k-1}(1-p_sp_m) + \sum_{i=1}^{n}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_sp_m)^{n-1}\right]}. \tag{20}$$

3.2. Expected maintenance cost in a renewal cycle

The maintenance cost in a renewal cycle includes the cost of inspections, repairs and a replacement. Accordingly, the expected maintenance cost in a renewal cycle is given by

$$E(C) = c_I E(N_I) + c_r E(N_f) + \tilde{c}_r. \tag{23}$$

where c_I, c_r and $\tilde{c}_r$ respectively denote the cost of an inspection, a repair and a replacement.

The expected number of inspections, $E(N_I)$, is given in (18). Using the probability mass function in (7), $E(N_f)$ is

$$E(N_f) = (1-p_sp_m)\sum_{k=1}^{n-1} k(p_sp_m)^{k-1} + n(p_sp_m)^{n-1} = \frac{1-(p_sp_m)^n}{1-p_sp_m}. \tag{24}$$

Substituting (18) and (24) into (23), the expected maintenance cost in a renewal cycle is expressed as

$$E(C) = c_I\left[\sum_{k=1}^{n-1}\sum_{i=1}^{k}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_sp_m)^{k-1}(1-p_sp_m) + \sum_{i=1}^{n}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_sp_m)^{n-1}\right] + c_r\left[\frac{1-(p_sp_m)^n}{1-p_sp_m}\right] + \tilde{c}_r. \tag{25}$$

Finally, the expected net revenue for the equipment supplier, which is a function of n and T, is obtained from (20) and (25) as follows:

$$\begin{aligned} R(T,n) = {} & a + b \\ & \times \left\{ \frac{(1-p_sp_m)\left(\sum_{k=1}^{n-1}(p_sp_m)^{k-1}\int_0^\infty tf_{S_k}(t)dt\right) + (p_sp_m)^{n-1}\int_0^\infty tf_{S_n}(t)dt}{T\left[\sum_{k=1}^{n-1}\sum_{i=1}^{k}\sum_{m=1}^{\infty} mP(N_I^i=m)(p_sp_m)^{k-1}(1-p_sp_m) + \sum_{i=1}^{n}\sum_{m=1}^{\infty} mP(N_I^i=m)(p_sp_m)^{n-1}\right]} - A_{\min} \right\} \\ & - \left\{ c_I\left[\sum_{k=1}^{n-1}\sum_{i=1}^{k}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_sp_m)^{k-1}(1-p_sp_m) \right.\right. \\ & \left.\left. + \sum_{i=1}^{n}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_sp_m)^{n-1}\right] + c_r\left[\frac{1-(p_sp_m)^n}{1-p_sp_m}\right] + \tilde{c}_r \right\}. \end{aligned} \tag{26}$$

Here, $P(N_I^i = m)$ is given in (16).

Remark. Under specific parameter settings, the proposed model can be transformed into several classic maintenance models.

Remark. For the special case of minimal repair, the probability mass function of N_I^i is given by

$$P(N_I^i = m) = \begin{cases} F(mT) - F((m-1)T), & i = 1 \\ \int_0^\infty \frac{[F(mT+x) - F((m-1)T+x)]\lambda(x)[H(x)]^{i-2}}{(i-2)!}dx, & i \geqslant 2 \end{cases} \tag{21}$$

where $F(x)$ is the distribution function of the system lifetime.

Specifically, for an exponential distribution with failure rate λ, we have

$$P(N_I^i = m) = e^{-\lambda(m-1)T} - e^{-\lambda mT}. \tag{22}$$

The derivations of (21) and (22) are given in Appendix A. The result in (22) coincides with our intuition that the exponential distribution is memory-less.

(1). When the probability of a soft failure is zero, that is $p_s = 0$, replacement is the only maintenance activity. Consequently, a preventive replacement policy is not applicable and only the inspection interval T needs to be determined. Such a model is appropriate for unrepairable systems such as the CPUs of servers and truck tyres. On the contrary, when $p_s = 1$, the system is repairable and a preventive replacement policy is applicable.

(2). When the probability of an imperfect repair is $p_m = 1$, no maintenance error occurs. In this case, the system is correctively replaced upon a hard failure or preventively replaced at the nth soft failure. This model was proposed by Sheu et al. (2015), and its object is to minimize the average long-run cost rate.

(3). When the preventive replacement limit $n = \infty$, only corrective replacement is available, and preventive replacement is no longer provided. Such a maintenance strategy is appropriate for those systems with a decreasing failure rate or a short life cycle such as seats on public transport.

4. Properties of optimal maintenance policies

Because the derivation of the net revenue function in (26) holds for general distributions, it is interesting to investigate the maintenance optimization for general distributions such as Weibull and Gamma distributions. Failure rate is expected to have a role in the analysis of the optimal maintenance policy (Gupta et al., 2017), therefore, we investigate the optimal maintenance policy from the perspective of system failure rate. When the failure rate of the system is *non-monotonic*, it may be difficult to obtain the analytical properties of the optimal maintenance policy. Nevertheless, for systems with *monotonic failure rate functions*, we have the following results.

4.1. Systems with monotonic non-increasing failure rate

For systems that do not age, i.e., systems with a constant or decreasing failure rate, preventive replacement is not a favorable strategy (Flage, 2014). Thus, the optimal solution can be expressed as $(T^*, n^*) = (T^*, \infty)$. Specifically, $n = \infty$ implies that the system is only correctively replaced when a maintenance error or hard failure occurs. We let $n = \infty$ in (26) and obtain

$$R(T,\infty) = a + b \times \left\{ \frac{(1-p_s p_m)\left(\sum_{k=1}^{\infty}(p_s p_m)^{k-1}\int_0^{\infty} t f_{S_k}(t)dt\right)}{T\left[\sum_{k=1}^{\infty}\sum_{i=1}^{k}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_s p_m)^{k-1}(1-p_s p_m)\right]} - A_{\min}\right\} - \left\{ c_I\left[\sum_{k=1}^{\infty}\sum_{i=1}^{k}\sum_{m=1}^{\infty} mP(N_I^i = m)(p_s p_m)^{k-1}(1-p_s p_m)\right] + \frac{c_r}{1-p_s p_m} + \tilde{c}_r \right\}. \quad (27)$$

Intuitively, the inspection interval should not be too small. Smaller T may reduce the down time cost; however, it can also increase the inspection cost. It may be difficult to obtain an analytical optimal inspection interval T^* directly from (27) for general systems with non-increasing failure rates. Hence, to obtain more insights about this optimal maintenance policy, we consider the extensively used exponential distribution with a constant failure rate.

Due to the important property of memoryless, exponential distribution is suitable for modelling units within the "bathtub curve," which is usually referred to as the useful life (Zhao, Chen, & Zhou, 2016). For systems with exponential distributions, we have the following theorem for determining the optimal maintenance policy.

Theorem 1. *When the failure time is exponentially distributed, the optimal maintenance policy can be expressed by (T^*, ∞), where T^* satisfies the following equation*

$$\frac{(1-e^{-\lambda T})^2[e^{-\lambda T}(\lambda T + 1) - 1]}{c_I \lambda^2} + \frac{T^2 e^{-\lambda T}}{b(1-p_s p_m)} = 0. \quad (28)$$

Proof. The proof of Theorem 1 is given in Appendix B. □

Remark. Theorem 1 states that the optimal inspection interval T^* under the exponential distribution exists and satisfies (28). Note that when the double derivative of $R(T,\infty)$ with respect to T is less than zero, then we can obtain a unique T^* by solving (28). Otherwise, we must determine whether the solution to (28) is a local or global optimal solution. A detailed numerical example for exponentially distributed system is presented in Section 5.2.

4.2. Systems with monotonically increasing failure rates

For systems with an *increasing failure rate function*, we introduce the following lemma.

Lemma 1. *When the failure rate of the system is an increasing function in time t, the expected operating time of the system between the $(n-1)$th and the nth imperfect repairs $E(X_n)$ is a decreasing function with respect to n.*

Proof. The proof of Lemma 1 is given in Appendix B. □

Remark. Lemma 1 conveys that for a system with an increasing failure rate, the successive operating time between failures monotonically decreases as the number of repairs increases. Obviously, this result coincides with the stochastic phenomenon that the working time of a deteriorating system after many repairs will continue to shorten. Using Lemma 1, the optimal maintenance policy under an increasing failure rate can be simplified, as shown in Theorem 2.

Theorem 2. *For systems with increasing failure rate, the optimal solution to (26) becomes $(T^*, n^*) = (T^*, 1)$, where T^* satisfies the following equation*

$$\frac{bE(X)[TM'(T) - M'(T)]}{[TM(T)]^2} - c_I M'(T) = 0, \quad (29)$$

where $M(T) = \sum_{m=1}^{\infty} m(F(mT) - F(m-1)T)$, and $M'(T) = dM(T)/dT$.

Proof. The proof of Theorem 2 is presented in Appendix B. □

Remark. Theorem 2 shows that given a fixed inspection interval, the steady-state availability $A(T,n)$ tends to decrease as n increases, whereas the maintenance cost increases. Therefore, the optimal solution is given by $(T^*, n^*) = (T^*, 1)$, i.e., replacing the system when the first failure occurs is the most economical way. Note that when the double derivative of $R(T,1)$ with respect to T is less than zero, then we can obtain a unique T^* by solving (29). Otherwise, we must determine whether the solution to (29) is a local or global optimal solution.

The above results show that when the system is subject to a non-monotonic failure rate, the optimal n^* could only be obtained numerically. Moreover, there are two decision variables in the maintenance model, i.e., T and n. In contrast, when the failure rate is monotonic, the value of n is fixed (at 1 for an increasing failure rate and ∞ for a decreasing failure rate) regardless of the parameter settings. Thus, there is only one decision variable T in the optimization model. In Section 5.3, the extensively adopted Weibull distribution with an increasing failure rate is chosen for illustration.

5. Numerical examples

5.1. Background

A Wind turbine system (WTS) is one type of system that converts wind power into electrical energy. Nowadays, WTSs play increasingly critical roles in meeting continuously increasing energy demands and achieving current the carbon emission reduction targets (Jafari & Makis, 2016; Shafiee, Finkelstein, & Bérenguer, 2015). A detailed description of WTSs can be found in Sun, Luo, and Wang (2015). In the past decade, PBC has emerged as a new service paradigm for the maintenance of WTSs (Jin et al., 2012). Specifically, in practice, WTS users determine the steady-state availability goal of the WTS and sign a contract with suppliers who are then committed to achieving this goal.

WTSs always operate in remote areas and it can be very costly to dispatch maintenance crews until a scheduled periodic inspection time. The inspection determines whether a WTS is operating normally or has had a soft or hard failure. If one of the failure modes has occurred, maintenance is performed. Otherwise, the system remains undisturbed until the next inspection. In this section, we numerically examine the trade-off between system maintenance cost and steady-state availability with the goal of maximizing the expected net revenue of the WTS supplier.

The parameters of the WTS are given as follows: (i) The fixed value and incentive coefficient of the contract are both USD 100000. To receive any compensation, the minimum steady-state availability the contractor needs to achieve is $A_{\min} = 0.5$. (ii) If a failure has occurred, the conditional probabilities of a soft and hard failure are respectively $p_s = 0.7$ and 0.3. When a soft failure occurs, the system is imperfectly repaired with probability $p_m = 0.8$ or replaced because of a maintenance error with probability $1 - p_m = 0.2$. (iii) The replacement cost of the system is $\tilde{c}_r =$ USD 10000, the cost of an inspection is $c_I =$ USD 500 and the cost of repair is $c_r =$ USD 1000.

In the following section, we present two numerical examples to illustrate the proposed model. In the first example, the lifetime of the system X is assumed to follow an exponential distribution and in the second example, X is assumed to follow a Weibull distribution.

5.2. Exponentially distributed X

If the lifetime of the system follows an exponential distribution with failure rate $\lambda = 0.5$, then, $F_X(t) = 1 - e^{-\lambda t}, t > 0$. The steady-state availability of the system, $A(T, \infty)$, can be obtained from (20) by letting $n \to \infty$. The results show that when T is small, increasing T may significantly decrease steady-state availability. However, as T increases further, the impact of T on steady-state availability decreases. When the inspection interval equals 3.2, the corresponding steady-state availability is 0.5. Note that $A_{\min} = 0.5$; thus, the inspection interval should be less than 3.2 to yield any benefit.

Fig. 1 shows the net-revenue for different inspection intervals. Generally, a smaller inspection interval will increase inspection expense, whereas a larger inspection interval will incur a higher system down time penalty cost. Therefore, an optimal interval should be chosen that balances the cost of inspections with that of system down time. This is illustrated in Fig. 1, where increasing T causes the net-revenue first to increase then to decrease. When $p_s = 0.7$ and $p_m = 0.8$, the optimal policy is to inspect the system every 0.35 time units, which yields $A(0.35, \infty) = 0.91$ and $R(0.35, \infty) =$ USD 122400. Additionally, according to Theorem 1, the optimal inspection interval can also be determined by solving (28).

In Fig. 1, we also show results for (i) $p_s = 0.7$, $p_m = 1$ and (ii) $p_s = 1$, $p_m = 0.8$. Note that $1 - p_s p_m$ denotes the conditional probability that the system is replaced given a failure has occurred. The results in Fig. 1 indicate that a lower value of $1 - p_s p_m$ corresponds to lower replacement costs and thus larger expected net revenue.

5.3. Weibull distributed X

In this section, we consider a system with Weibull distribution. The probability density function of the Weibull distribution with shape parameter α and scale parameter λ is given by

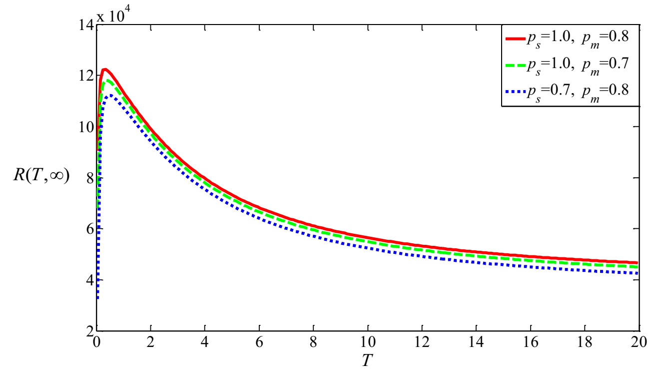

Fig. 1. Expected net revenue versus inspection interval for different probabilities.

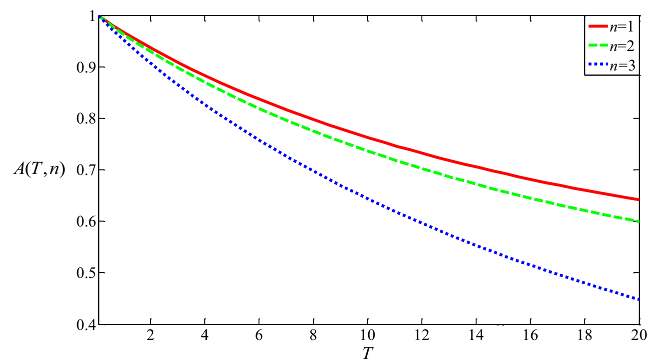

Fig. 2. Steady-state availability versus T and n.

$$f(t) = \frac{\alpha}{\lambda}\left(\frac{t}{\lambda}\right)^{\alpha-1} e^{-\left(\frac{t}{\lambda}\right)^{\alpha}}, t > 0, \alpha > 0, \lambda > 0.$$

Correspondingly, the failure rate function of the system is $\lambda(t) = (\alpha/\lambda)(x/\lambda)^{\alpha-1}$, and the hazard function is $H(t) = (t/\lambda)^{\beta}$. When $\alpha > 1$, the Weibull distribution failure rate increases. In this example, the parameters of the distribution are $\lambda = 10$ and $\alpha = 1.3$. The steady-state availability versus T and n is obtained using (20) and is shown in Fig. 2. We can see that given a fixed n, steady-state availability $A(T, n)$ decreases as T increases. If the system is inspected more frequently, failures will be detected and addressed in time, resulting in the increase of the steady-state availability. The proof of Theorem 2 shows that, for a given fixed inspection interval T, steady-state availability $A(T, n)$ decreases as n increases, as illustrated by Fig. 2.

The expected net revenue in terms of T for selected n is presented in Fig. 3. Given a fixed T, $R(T, n)$ decreases as n increases,

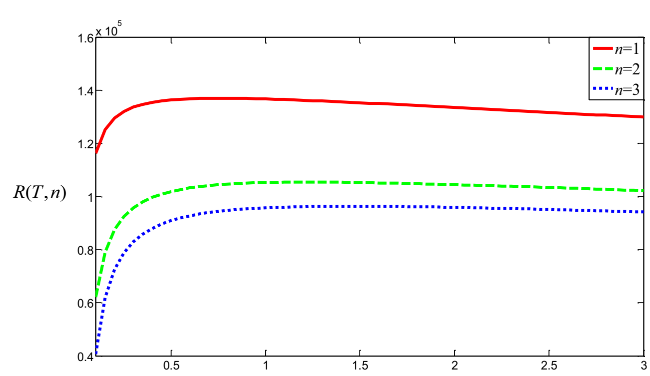

Fig. 3. Expected net revenue with respect to T under three different values of n.

hence, we have $R(T,1) > R(T,n), n = 2,3,\ldots$. If n is fixed, expected net revenue $R(T,n)$ increases first and then roughly decreases with respect to T, i.e., there exists an optimal inspection interval maximizing the net-revenue. Theorem 2 can be employed to find the optimal policy. The optimal maintenance policy is to inspect the system each 0.7 time units and preventively replace the system at the first failure, which yields $R(0.7,1) =$ USD 136900. From Fig. 2, the corresponding steady-state availability is $A(0.7,1) = 0.93$.

6. Conclusions

This paper investigated the optimal maintenance policy for inspected systems that are subject to both soft and hard failures and operating under a PBC. In this policy, maintenance errors are also incorporated into the model formulation. Several classic maintenance models are shown to be special cases of our model. Analytical expressions for the expected net revenue of suppliers were derived using the Kijima virtual age model. These expressions were used to analyse some essential properties of the optimal solution. The proposed maintenance model was illustrated by a numerical WTS example.

The current study could be extended along several lines. It would be of interest to investigate the optimal maintenance policy for multiple failure-mode systems with imperfect inspections (Flage, 2014; Ye, Chen, & Tsui, 2015). Because of the rapid development of sensing technologies, a condition-based inspection schedule (Yang, Ma, & Zhao, 2017) should be more effective but more complex. Furthermore, the independent failure modes considered in this paper could be extended to a more realistic dependent case (Liu et al., 2013; Zhang et al., 2016).

Additionally, we could consider the Bayesian imperfect repair model (Lim, Qu, & Zuo, 2016) in which the probabilities of different repairs may not be predetermined and are chosen as random variables with certain distributions. Finally, age based analysis alone is not sufficient and may not reflect the true system reliability. In this context, the maintenance policy may also depend on the distribution of the usage. It is of both theoretical and practical interest to address maintenance that is conditional on usage (Gupta et al., 2017).

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Appendix A

Derivation of (13)

From (10), for the exponential distribution, the expected uptime in a renewal cycle is derived as follows

$$\begin{aligned}E(U) &= (1-p_sp_m)\left(\sum_{k=1}^{n-1}(p_sp_m)^{k-1}\int_0^\infty \frac{tf(t)[H(t)]^{k-1}}{(k-1)!}dt\right) + (p_sp_m)^{n-1} \\ &\quad\times \int_0^\infty \frac{tf(t)[H(t)]^{n-1}}{(n-1)!}dt \\ &= (1-p_sp_m)\left(\sum_{k=1}^{n-1}(p_sp_m)^{k-1}\int_0^\infty \frac{t\lambda e^{-\lambda t}(\lambda t)^{k-1}}{(k-1)!}dt\right) + (p_sp_m)^{n-1} \\ &\quad\times \int_0^\infty \frac{t\lambda e^{-\lambda t}(\lambda t)^{k-1}}{(n-1)!}dt \\ &= \frac{1-(p_sp_m)^{n-1}+(1-p_sp_m)(p_sp_m)^{n-1}}{\lambda(1-p_sp_m)}.\end{aligned}$$

Derivation of Eq. (18)

Using the total probability decomposition method, the expected number of inspections in a renewal cycle $E(N_I)$ can be expressed as

$$\begin{aligned}E(N_I) &= E\left(\sum_{i=1}^{N_f}N_I^i\right) = \sum_{k=1}^{n-1}E\left(\sum_{i=1}^{k}N_I^i\right)P(N_f=k) + E\left(\sum_{i=1}^{n}N_I^i\right)P(N_f=n) \\ &= \sum_{k=1}^{n-1}\sum_{i=1}^{k}\sum_{m=1}^{\infty}mP(N_I^i=m)(p_sp_m)^{k-1}(1-p_sp_m) \\ &\quad+\sum_{i=1}^{n}\sum_{m=1}^{\infty}mP(N_I^i=m)(p_sp_m)^{n-1}.\end{aligned}$$

Derivation of Eq. (21)

The probability mass function of N_I^i is expressed as

$$P(N_I^i=m) = P((m-1)T < X_i < mT).$$

For $i=1$,

$$P(N_I^1=m) = P((m-1)T < X_1 < mT) = F(mT) - F((m-1)T).$$

For $i \geqslant 2$,

$$\begin{aligned}P(N_I^i=m) &= P((m-1)T < X_i < mT) = \int_0^\infty P((m-1)T < X_i \\ &\quad < mT|S_{i-1}\in(x,x+\Delta x))dF_{S_{i-1}}(x) \\ &= \int_0^\infty \frac{[F(mT+x)-F((m-1)T+x)]\lambda(x)[H(x)]^{i-2}}{(i-2)!}dx.\end{aligned}$$

Hence the probability mass function of N_I^i is obtained as

$$P(N_I^i=m) = \begin{cases}F(mT)-F((m-1)T), & i=1\\ \int_0^\infty \frac{[F(mT+x)-F((m-1)T+x)]\lambda(x)[H(x)]^{i-2}}{(i-2)!}dx, & i\geqslant 2\end{cases}$$

Derivation of Eq. (22)

For an exponential distribution with failure rate λ, the number of inspections can be calculated by Eq. (21).

For $i=1$,

$$P(N_I^i=m) = F(mT)-F((m-1)T) = e^{-\lambda(m-1)T}-e^{-\lambda mT}.$$

For $i \geqslant 2$, we obtain

$$\begin{aligned}P(N_I^i=m) &= \int_0^\infty \frac{[F(mT+x)-F((m-1)\tau+x)]\lambda(x)[H(x)]^{i-2}}{(i-2)!}dx \\ &= \int_0^\infty \frac{[e^{-\lambda[(m-1)T+x]}-e^{-\lambda(mT+x)}]\lambda(\lambda x)^{i-2}}{(i-2)!}dx = e^{-\lambda(m-1)T}-e^{-\lambda mT}.\end{aligned}$$

Hence, for the exponential distribution we have

$$P(N_I^i=m) = e^{-\lambda(m-1)T}-e^{-\lambda mT}.$$

Appendix B

Proof of Theorem 1. Using the result in (27), for an exponential distribution, the objective function $R(T,\infty)$ is given by

$$\begin{aligned}&R(T,\infty) \\ &= a+b\times\left\{\frac{\sum_{k=1}^{\infty}k(p_sp_m)^{k-1}}{\lambda T\left[\sum_{k=1}^{\infty}\sum_{i=1}^{k}\sum_{m=1}^{\infty}m[e^{-\lambda(m-1)T}-e^{-\lambda mT}](p_sp_m)^{k-1}\right]}-A_{\min}\right\} \\ &\quad-\left\{c_I\left[\sum_{k=1}^{\infty}\sum_{i=1}^{k}\sum_{m=1}^{\infty}m[e^{-\lambda(m-1)T}-e^{-\lambda mT}](p_sp_m)^{k-1}(1-p_sp_m)\right]\right. \\ &\quad\left.+\frac{c_r}{1-p_sp_m}+\tilde{c}_r\right\},\end{aligned}$$

which can be simplified as

$$R(T,\infty) = a + b \times \left\{ \frac{1-e^{-\lambda T}}{\lambda T} - A_{\min} \right\} - \left\{ \frac{c_I}{(1-e^{-\lambda T})(1-p_s p_m)} + \frac{c_r}{1-p_s p_m} + \tilde{c}_r \right\}.$$

Differentiating $R(T,\infty)$ with respect to T, we obtain that $dR(T,\infty)/dT = 0$ holds if and only if

$$\frac{(1-e^{-\lambda T})^2 [e^{-\lambda T}(\lambda T + 1) - 1]}{c_I \lambda^2} + \frac{T^2 e^{-\lambda T}}{b(1-p_s p_m)} = 0.$$

Thus, the optimal inspection interval exists and satisfies the following equation

$$\frac{(1-e^{-\lambda T})^2 [e^{-\lambda T}(\lambda T + 1) - 1]}{c_I \lambda^2} + \frac{T^2 e^{-\lambda T}}{b(1-p_s p_m)} = 0. \quad \square$$

Proof of Lemma 1. Suppose that the survival function of S_n is $R_n(x)$, which can be derived using the total probability decomposition method as

$$R_n(x) = P(S_n > x) = R_{n-1}(x) + \int_0^x \frac{R\left(y + \frac{x-y}{a}\right)}{R(y)} dF_{n-1}(y).$$

Then $E(X_n)$ is expressed as

$$E(X_n) = \int_0^\infty [R_n(x) - R_{n-1}(x)]dx = \int_0^\infty \int_0^x \frac{R\left(y + \frac{x-y}{a}\right)}{R(y)} dF_{n-1}(y)dx.$$

When $\lambda(t)$ is an increasing function in time t, we have that $R(y + \frac{x-y}{a})/R(y)$ decreases in y. From Cui, Kuo, Loh, and Xie (2004), the following inequality holds

$$\int_0^\infty \int_0^x \frac{R\left(y + \frac{x-y}{a}\right)}{R(y)} dF_{n-1}(y)dx > \int_0^\infty \int_0^x \frac{R\left(y + \frac{x-y}{a}\right)}{R(y)} dF_n(y)dx,$$

which implies that $E(X_n) > E(X_{n+1})$. Therefore, $E(X_n)$ is a decreasing function with respect to n. $\square$

Proof of Theorem 2. The expected uptime and length of a renewal cycle are respectively a function of n and denoted by $E(U_n)$ and $E(L_n)$. According to Lemma 1, we have the following inequality

$$\frac{E(U_{n+1}) - E(U_n)}{E(L_{n+1}) - E(L_n)} < \frac{E(U_n)}{E(L_n)},$$

which implies

$$A(T, n+1) = \frac{E(U_{n+1})}{E(L_{n+1})} < \frac{E(U_n)}{E(L_n)} = A(T,n).$$

Obviously, the maintenance cost increases as n increases, therefore, $C(T,n) < C(T,n+1)$. Moreover $R(T,n)$ can be expressed as

$$R(T,n) = a + b[A(T,n) - A_{\min}] - C(T,n),$$

Combining the two inequalities, we can show that $R(T,n)$ is a decreasing function of n. Hence, the optimal solution is given by $(T^*, n^*) = (T^*, 1)$. The expected net revenue in (26) can be simplified as

$$R(T,1) = a + b \times \left\{ \frac{E(X)}{T\sum_{m=1}^\infty m(F(mT) - F(m-1)T)} - A_{\min} \right\} - \left\{ c_I \sum_{m=1}^\infty m(F(mT) - F(m-1)T) + c_r + \tilde{c}_r \right\}. \quad (30)$$

Suppose that $M(T) = \sum_{m=1}^\infty m(F(mT) - F(m-1)T)$ and the first derivative of $M(T)$ is denoted by $M'(T)$. To obtain the analytical result of the optimal solution, we calculate the first derivative of the objective function in (30) as

$$R'(T,1) = \frac{bE(X)[TM'(T) - M(T)]}{[TM(T)]^2} - c_I M'(T).$$

Thus optimal inspection interval exists and satisfies the following equation

$$\frac{bE(X)[TM'(T) - M(T)]}{[TM(T)]^2} - c_I M'(T) = 0. \quad \square$$

References

Berkowitz, D., Gupta, J. N. D., Simpson, J. T., & Mcwilliams, J. B. (2005). Defining and implementing performance based logistics in government. *Defense Acquisition Review Journal, 37*, 255–267.

Caputo, A. C., Pelagagge, P. M., & Salini, P. (2017). Modeling human errors and quality issues in kitting processes for assembly lines feeding. *Computers & Industrial Engineering, 111*, 492–506.

Cha, J. H., Lee, S., & Mi, J. (2004). Bounding the optimal burn-in time for a system with two types of failure. *Naval Research Logistics, 51*, 1090–1101.

Cheng, G. Q., & Li, L. (2012). A geometric process repair model with inspections and its optimisation. *International Journal of Systems Science, 43*(9), 1650–1655.

Cui, L., Kuo, W., Loh, H. T., & Xie, M. (2004). Optimal allocation of minimal & perfect repairs under resource constraints. *IEEE Transactions on Reliability, 53*(2), 193–199.

Cui, L., & Xie, M. (2001). Availability analysis of periodically inspected systems with random walk model. *Journal of Applied Probability, 38*, 860–871.

Cui, L., & Xie, M. (2005). Availability of a periodically inspected system with random repair or replacement times. *Journal of Statistical Planning and Inference, 131*, 89–100.

Devries, H. J. (2004). Performance based logistics-barriers and enablers to effective implementation. *Defense Acquisition Review Journal, 11*, 243–254.

Doyen, L., & Gaudoin, O. (2004). Classes of imperfect repair models based on reduction of failure intensity or virtual age. *Reliability Engineering & System Safety, 84*(1), 45–56.

Flage, R. (2014). A delay time model with imperfect and failure-inducing inspections. *Reliability Engineering and System Safety, 124*, 1–12.

Golmakani, H. R., & Moakedi, H. (2012). Periodic inspection optimization model for a two-component repairable system with failure interaction. *Computers & Industrial Engineering, 63*(3), 540–545.

Gupta, S. K., De, S., & Chatterjee, A. (2017). Some reliability issues for incomplete two-dimensional warranty claims data. *Reliability Engineering & System Safety, 157*, 64–77.

Jafari, L., & Makis, V. (2016). Joint optimization of lot-sizing and maintenance policy for a partially observable two-unit system. *The International Journal of Advanced Manufacturing Technology, 87*(5–8), 1621–1639.

Jin, T., Ding, Y., Guo, H., & Nalajala, N. (2012). Managing wind turbine reliability and maintenance via performance-based contract. *Power and Energy Society General Meeting*, 1–6.

Khatab, A., Ait-Kadi, D., & Rezg, N. (2014). Availability optimisation for stochastic degrading systems under imperfect preventive maintenance. *International Journal of Production Research, 52*(14), 4132–4141.

Kijima, M. (1989). Some results for repairable systems with general repair. *Journal of Applied probability, 26*(1), 89–102.

Levitin, G., Zhang, T., & Xie, M. (2006). State probability of a series-parallel repairable system with two-types of failure modes. *International Journal of Systems Science, 37*, 1011–1020.

Lim, J. H., Qu, J., & Zuo, M. J. (2016). Age replacement policy based on imperfect repair with random probability. *Reliability Engineering & System Safety, 149*, 24–33.

Lin, Y. K., Lin, J. J., & Yeh, R. H. (2013). A dominant maintenance strategy assessment model for localized third-party logistics service under performance-based consideration. *Quality Technology and Quantitative Management, 10*(2), 221–240.

Liu, X., Li, J., Al-Khalifa, K. N., Hamouda, A. S., Coit, D. W., & Elsayed, E. A. (2013). Condition-based maintenance for continuously monitored degrading systems with multiple failure modes. *IIE Transactions, 45*(4), 422–435.

Lu, B., Zhou, X., & Li, Y. (2016). Joint modeling of preventive maintenance and quality improvement for deteriorating single-machine manufacturing systems. *Computers & Industrial Engineering, 91*, 188–196.

Mirzahosseinian, H., & Piplani, R. (2011). A study of repairable parts inventory system operating under performance-based contract. *European Journal of Operational Research, 214*(2), 256–261.

Ng, I. C. L., Maull, R., & Yip, N. (2009). Outcome-based contracts as a driver for systems thinking and service-dominant logic in service science: Evidence from the defence industry. *European Management Journal, 27*, 377–387.

Nowicki, D., Kumar, U. D., Steudel, H. J., & Verma, D. (2008). Spares provisioning under performance-based logistics contract: Profit-centric approach. *Journal of the Operational Research Society, 59*(3), 342–352.

Peng, G. (2014). On human errors in maintenance: Risk potential and mitigation (Master's thesis). Norway: University of Stavanger.

Peng, H., Feng, Q., & Coit, D. W. (2010). Reliability and maintenance modeling for systems subject to multiple dependent competing failure processes. *IIE Transactions, 43*(1), 12–22.

Qiu, Q., Cui, L., & Gao, H. (2017). Availability and maintenance modelling for systems subject to multiple failure modes. *Computers & Industrial Engineering, 108*, 192–198.

Rankin, W. L. (2000). The maintenance error decision aid (MEDA) process. *Proceedings of the human factors and ergonomics society annual meeting* (Vol. 44, pp. 795–798). .

Ross, S. M. (1996). *Stochastic processes*. New York: John Wiley & Sons.

Selviaridis, K., & Wynstra, F. (2015). Performance-based contracting: A literature review and future research directions. *International Journal of Production Research, 53*(12), 3505–3540.

Shafiee, M., Finkelstein, M., & Bérenguer, C. (2015). An opportunistic condition-based maintenance policy for offshore wind turbine blades subjected to degradation and environmental shocks. *Reliability Engineering & System Safety, 142*, 463–471.

Sheu, S. H., Li, S. H., & Chang, C. C. (2012). A generalised maintenance policy with age dependent minimal repair cost for a system subject to shocks under periodic overhaul. *International Journal of System Science, 43*, 1007–1013.

Sheu, S. H., Tsai, H. N., Wang, F. K., & Zhang, Z. G. (2015). An extended optimal replacement model for a deteriorating system with inspections. *Reliability Engineering & System Safety, 139*, 33–49.

Sun, H., Luo, X., & Wang, J. (2015). Feasibility study of a hybrid wind turbine system–Integration with compressed air energy storage. *Applied Energy, 137*, 617–628.

Taghipour, S., Banjevic, D., & Jardine, A. K. S. (2010). Periodic inspection optimization model for a complex repairable system. *Reliability Engineering and System Safety, 95*(9), 944–952.

Xiao, H., & Peng, R. (2014). Optimal allocation and maintenance of multi-state elements in series–parallel systems with common bus performance sharing. *Computers & Industrial Engineering, 72*, 143–151.

Yang, L., Ma, X., & Zhao, Y. (2017). A condition-based maintenance model for a three-state system subject to degradation and environmental shocks. *Computers & Industrial Engineering, 105*, 210–222.

Ye, Z., Chen, N., & Tsui, K. L. (2015). A Bayesian approach to condition monitoring with imperfect inspections. *Quality and Reliability Engineering International, 31*(3), 513–522.

Zhang, J., Huang, X., Fang, Y., Zhou, J., Zhang, H., & Li, J. (2016). Optimal inspection-based preventive maintenance policy for three-state mechanical components under competing failure modes. *Reliability Engineering and System Safety, 152*, 95–103.

Zhao, L., Chen, M., & Zhou, D. (2016). General (N, T, τ) opportunistic maintenance for multicomponent systems with evident and hidden failures. *IEEE Transactions on Reliability, 65*(3), 1298–1313.

Zhao, X., & Xie, M. (2017). Using accelerated life tests data to predict warranty cost under imperfect repair. *Computers & Industrial Engineering, 107*, 223–234.

Zheng, Z., Zhou, W., Zheng, Y., & Wu, Y. (2016). Optimal maintenance policy for a system with preventive repair and two types of failures. *Computers & Industrial Engineering, 98*, 102–112.